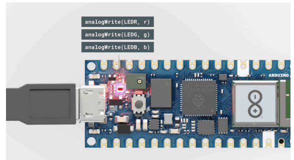
This tutorial refers to a product that has reached its end-of-life status.

Setting up Nano RP2040 Connect with Arduino Cloud

Learn how to access the IMU data and control the built-in RGB via the Arduino Cloud.

Author · Karl Söderby · Last revision · 01/05/2024



Nano RP2040 Connect.

Introduction

In this tutorial, we will go through the steps needed to connect the Nano RP2040 Connect to the Arduino Cloud.

We will test the connection out, by viewing real time data of the IMU (Inertial Measurement Unit), and remotely controlling the RGB on the board.

Goals

The goals of this project are:

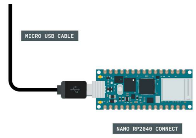
- Set up the Arduino Cloud.
- Read IMU sensor data.
- Control the built-in RGB.

Hardware & Software Needed

- Arduino Cloud.
- Arduino Nano RP2040 Connect.

Circuit

Follow the circuit below to connect the buttons and LEDs to your Arduino board.



This tutorial requires no additional circuit.

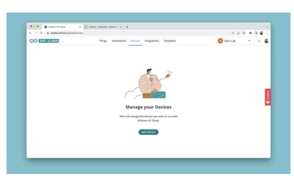
The Arduino Cloud

To start, we will need to head over to the **Arduino Cloud**.

Step 1: Configure Your Device

Once we are in the IoT Cloud, to configure our device, click on the **"Devices"** tab. Click on **"Add Device"** and follow the configuration to set up your board.

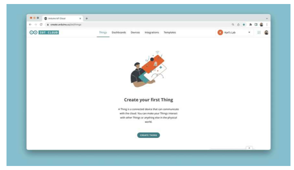
> Make sure your device is connected to your computer via USB at this point.



Configure a device.

Step 2: Create a Thing

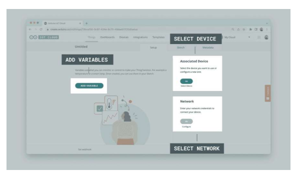
The next step is to create a new Thing. Navigate to the **"Things"** tab, and click on **"Create Thing".**



Create a Thing.

When you create a Thing, you will see a number of options:

- **Select Device** - the device you configured can be selected here.
- **Select Network -** enter your network credentials (Wi-Fi name and password).
- **Add Variables** - here we add variables that we will synchronize with our sketch.

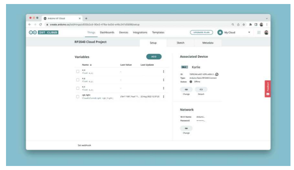


Thing overview.

The variables we will add are for controlling the **RGB** and to reading the **Accelerometer**. You can create them based on the table below:

Variable Name	Data Type	Permission
a_x	float	read only
a_y	float	read only
a_z	float	read only
rgb_light	Colored Light	read & write

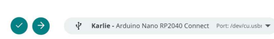
Once the variables are created, your Thing overview should look similar to the image below:



The final view.

Step 3: The Sketch

Now, we need to create the program for our Thing. First, let's head over to the **"Sketch"** tab in the Arduino Cloud. At the top, your board should be available in the dropdown menu by default.



Nano RP2040 Connect from board list.

The code can be found in the snippet below. Upload the code to your board.

```
#include "thingProper
#include <Arduino_LSM

void setup() {

  Serial.begin(9600);
  delay(1500);

  //define RGB LED as
  pinMode(LEDR, OUTPU
  pinMode(LEDG, OUTPU
  pinMode(LEDB, OUTPU

  initProperties();

  ArduinoCloud.begin(

  setDebugMessageLeve
  ArduinoCloud.printD

  //init IMU library
  if (!IMU.begin()) {
    Serial.println("F
    while (1);
  }
}

void loop() {
```

Once the code has been uploaded to your board, it will attempt to connect to your network, and then with the Arduino Cloud. You can open the Serial Monitor to see if there are any errors.

If all went well, you will see a message like this:

```
***** Arduino Cloud - c
Device ID: <your-device
Thing ID: <your-thing-i
MQTT Broker: mqtts-sa.i
WiFi.status(): 0
Current WiFi Firmware:
Connected to "<your-net
Connected to Arduino Cl
```

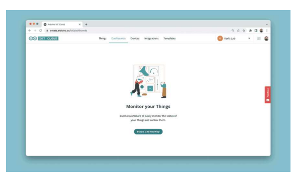
If it went wrong, you may encounter errors such as:

```
Connection to "<your-ne
```

Which indicate that something might be wrong with your network credentials.

Step 4: Creating a Dashboard

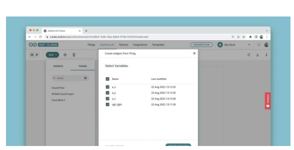
Once our sketch is successfully uploaded we can finalize this project by creating a dashboard. To do this, we need to navigate to the **"Dashboards"** tab, and click on the **"Build Dashboard"** tab.



Create a Dashboard.

This will open an empty dashboard. At the top left corner, first click on the **Pen Symbol**. You are now in Edit Mode.

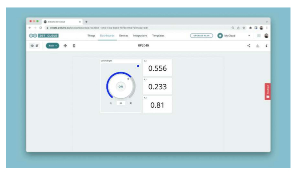
Now, click on the **"ADD"** button, then **"Things"**, search and click on your Thing. You will now see a list of variables with a checkmark. Leave all marked, and click on the **"Create Widgets** button.



Click on the "Edit button", the pen symbol.

We should now have a pretty nice looking dashboard that displays the value of our accelerometer, and a **colored light widget** to control the built-in RGB on the Nano RP2040 Connect.

These widgets can be re-sized and adjusted to your liking. You can also select and link other types of widgets, but this is by far the quickest option.

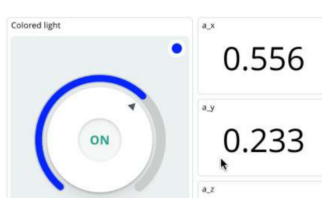


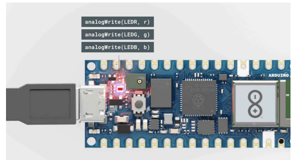
Final look of the dashboard.

Expected Outcome

Congratulations. You have now configured your Nano RP2040 Connect board with the Arduino Cloud service.

If your board successfully connects to the Arduino Cloud, we should be seeing values in the dashboard update continuously, and we can change the color of the RGB through the colored light widget.





Controlling the RGB on the Nano RP2040 Connect.

Conclusion

In this tutorial, we connected a Nano RP2040 Connect board to the **Arduino Cloud**. We created a simple sketch that allows us to display IMU sensor data directly in the Arduino Cloud dashboard, and how to control the built-in RGB on the board.

More Tutorials

For more interesting tutorials around the IoT Cloud, check out the **Arduino Cloud documentation page**.

To learn more about the Nano RP2040 Connect board, you can check out the **Arduino Nano RP2040 Connect documentation page**.

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anything wrong, you can edit this page here.

Was this article helpful?